

LLM Lecture Notes

Wednesday, October 22, 2025 9:05 PM

Summary of Attention:

- Prior to attention was RNN

- Attention operated in parallel which was much faster

- Attention allowed LLMs to function significantly better

Billions of parameters make up a single LLM

Trained on thousands of years of human experience

Continue to get bigger as time goes on

AI can beat Olympic level math problems

- Mathematic proofs have concrete steps

Scale:

- GPT 3 had 96 attention heads per layer

GPTs only have decoders, they don't have encoders

Cost to train BERT:

- Thousands of dollars

Some cost millions

- Test trains on smaller models to see if they work well

Bidirectional would give better sentence embeddings

- LLMs have to go left to right, too expensive to go bidirectional

- Bidirectional same size as a left to right model would be better

Predicting left to right

- As we add words they do attention on the words behind them

- Add a word to sentence, run again to get next word and next word and so on

GPT stands for pre-trained

- Early didn't have an interactive chat style

Zero shot

- No example

Few shot

- Give examples before the prompt

After pre-trained we do supervised fine-tuning

- Humans interact with chatbots and fine-tune the generated results

Gandalf llm - game

- Trying to get the LLM to give you the password

After fine-tuning it goes through reinforcement learning with human supervision

- Model generates different outputs

- Human ranks the outputs

- Can lead to models trending towards what sounds good or looks good

Hallucination:

Gives answers that sound good, even if they're not completely true
Tend to complement the user a lot

Train to give out cold hard facts:
Easier to do with math

Makeup test on Monday
Group projects still due Tuesday night